AmBe Thermal Neutron Coincidence Analysis

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SUMMARY STATISTICS
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Dataset Information:
  • Total events analyzed: 11,098
  • Date generated: 2026-03-26 16:23:13
  • Data source: two_channel_all_events.csv

Channel 1 (Gamma) Statistics:
  • Amplitude (V):     mean = 0.9485 ± 0.7829
                       range = [0.1494, 2.3652]

  • Charge (V·s):      mean = 1.219e-08 ± 1.220e-08
                       range = [1.043e-09, 1.041e-07]

Channel 2 (Neutron) Statistics:
  • Amplitude (V):     mean = 0.1668 ± 0.1393
                       range = [-0.0000, 0.3967]

  • Charge (V·s):      mean = 2.605e-09 ± 3.386e-09
                       range = [-1.121e-08, 1.674e-08]

  • Saturated events: 1967 (17.7%)

Time-of-Flight (Coincidence):
  • Delta-t (ns):      mean = 0.12 ± 34.36
                       range = [-127.16, 115.05]
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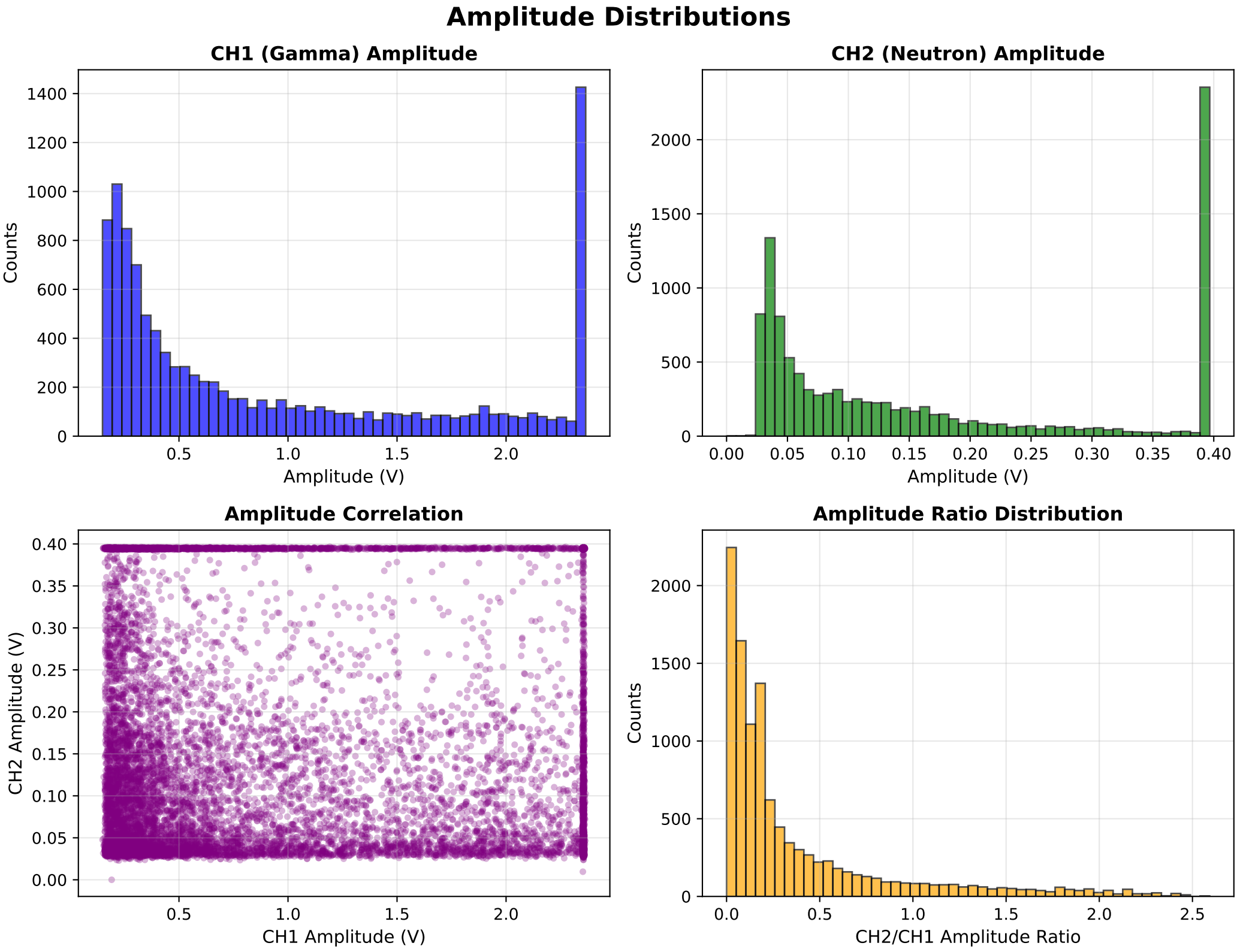
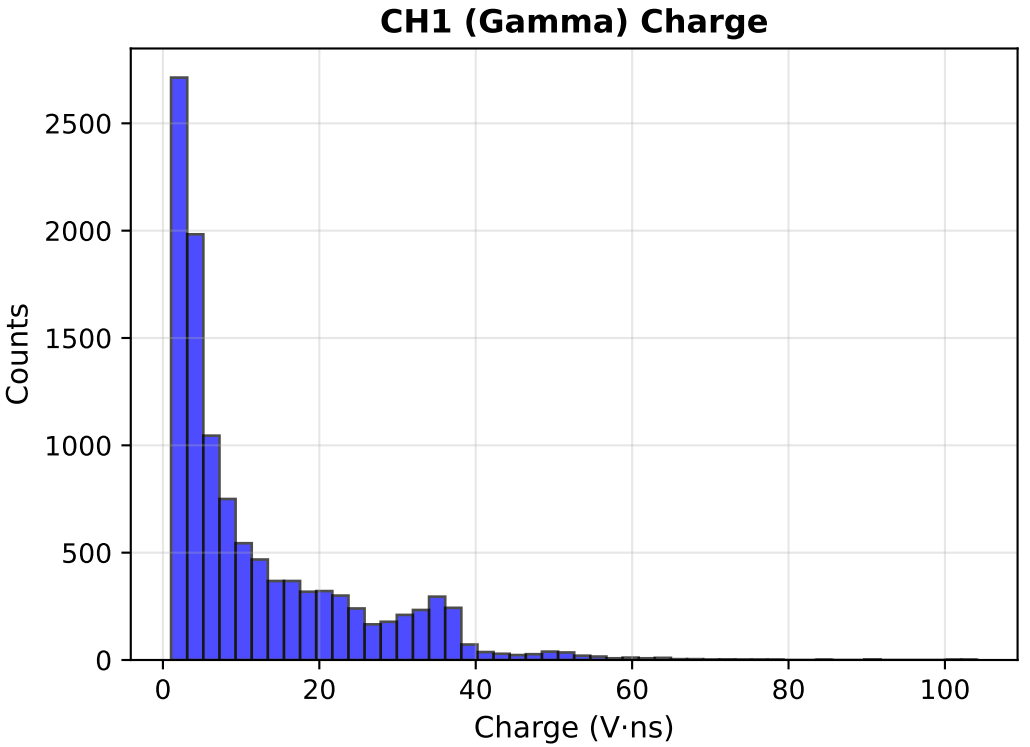
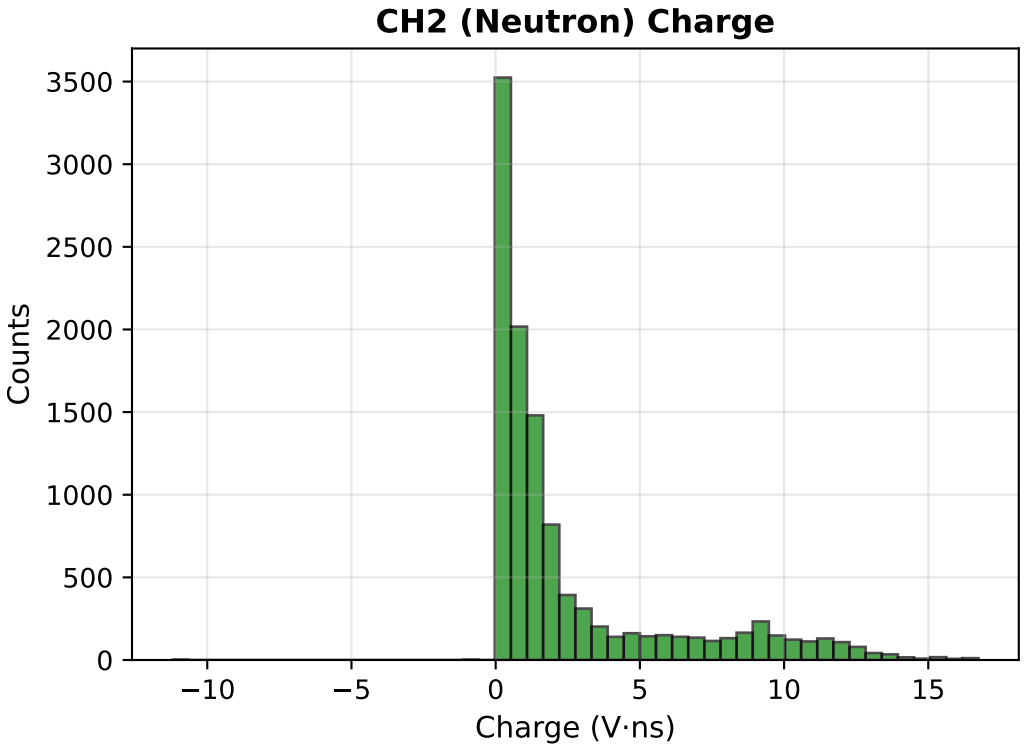
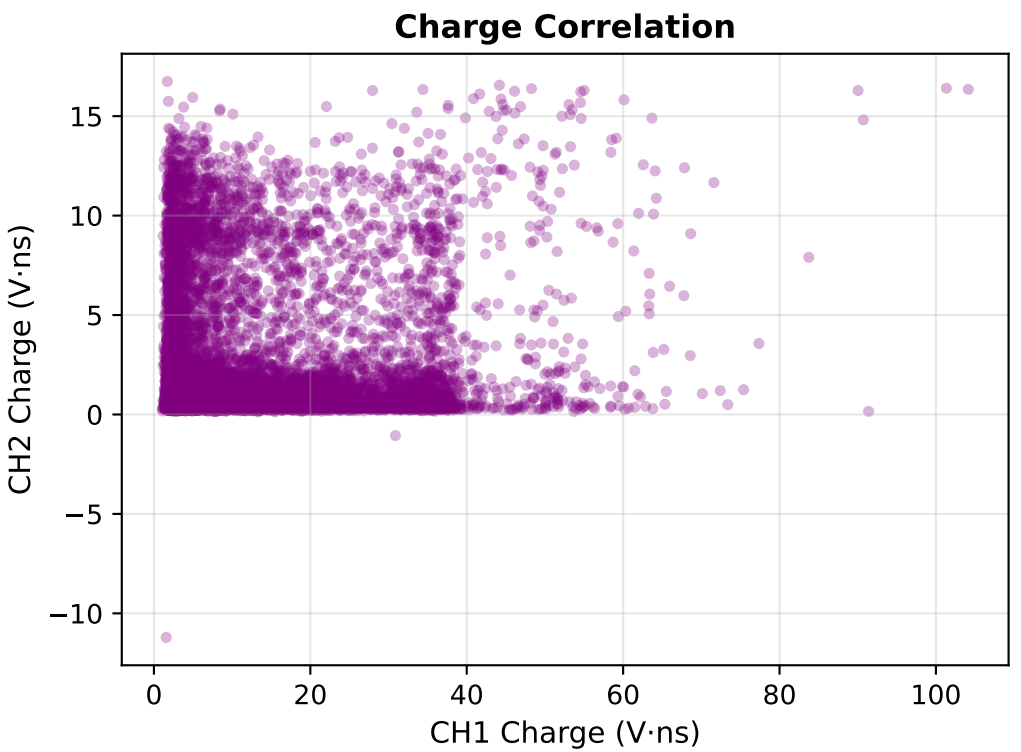
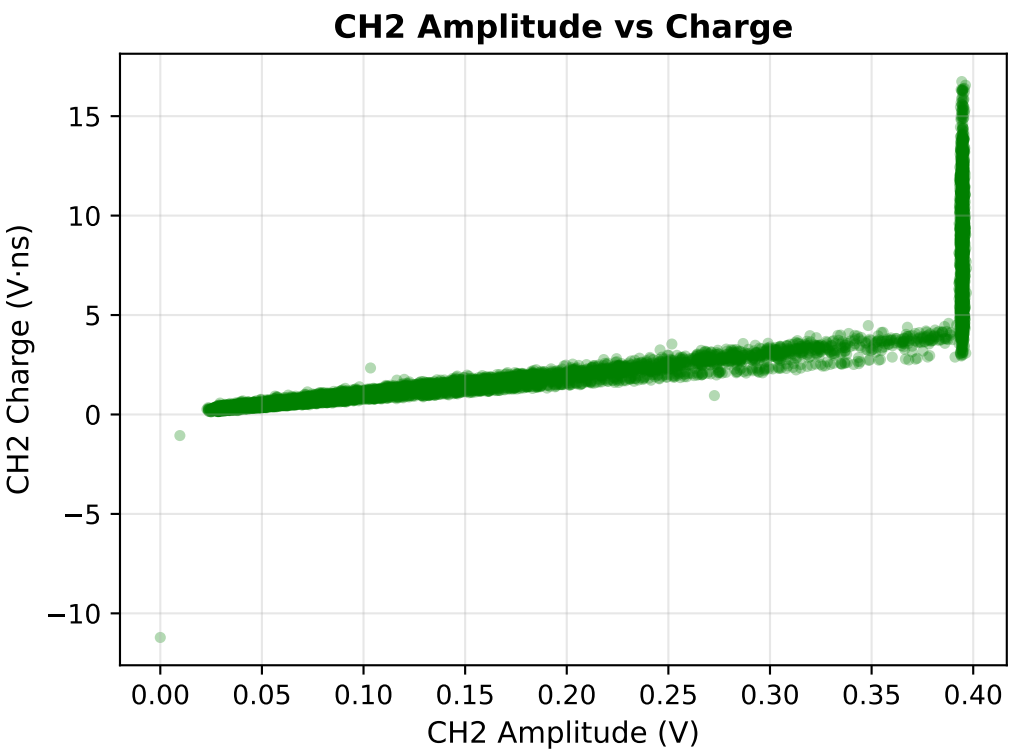

Amplitude Distributions
CH1 (Gamma) Amplitude
Counts
Amplitude (V)
CH2 (Neutron) Amplitude
Counts
Amplitude (V)
Amplitude Correlation
CH2 Amplitude (V)
CH1 Amplitude (V)
Amplitude Ratio Distribution
Counts
CH2/CH1 Amplitude Ratio

Integrated Charge Distributions

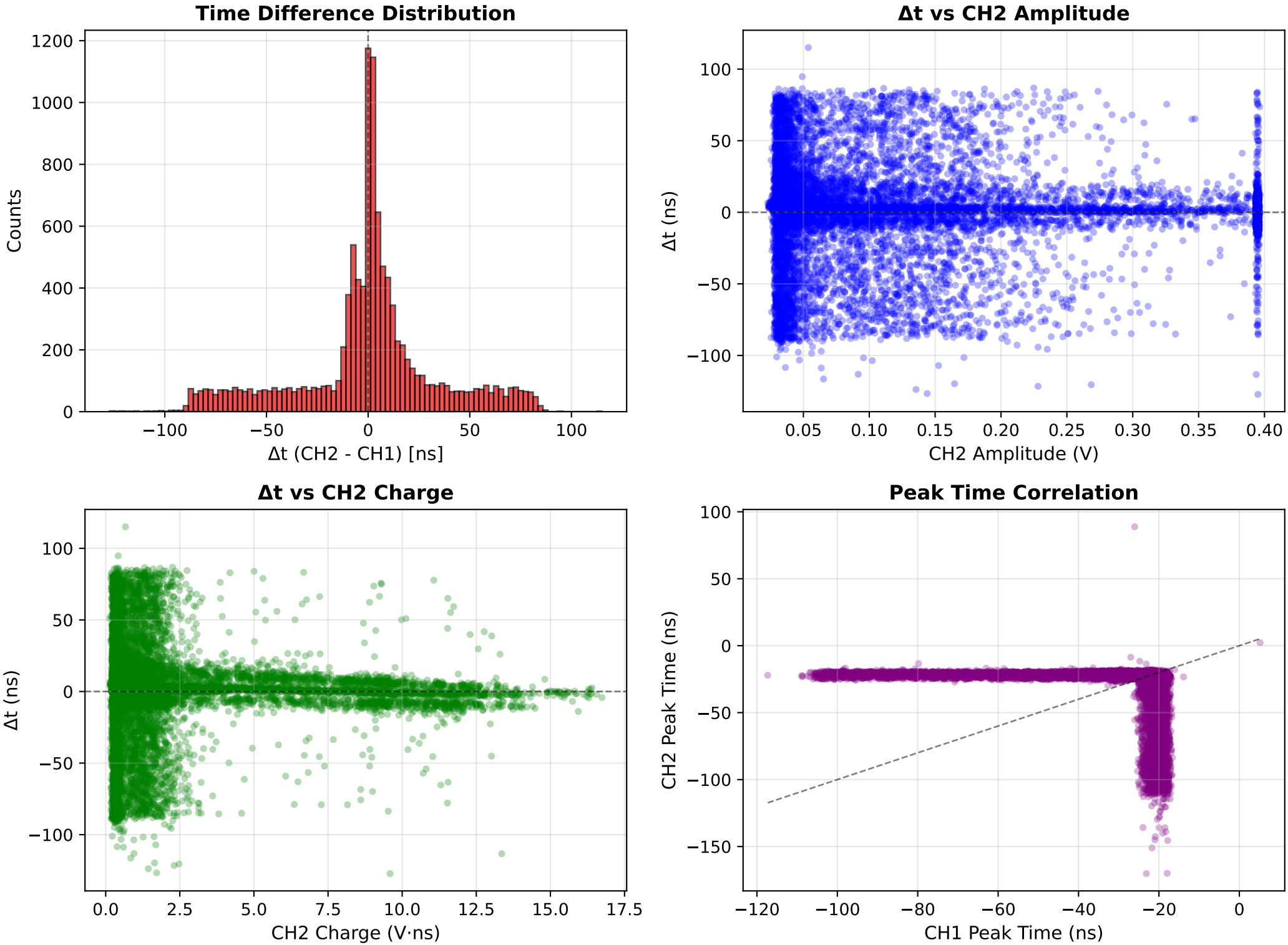
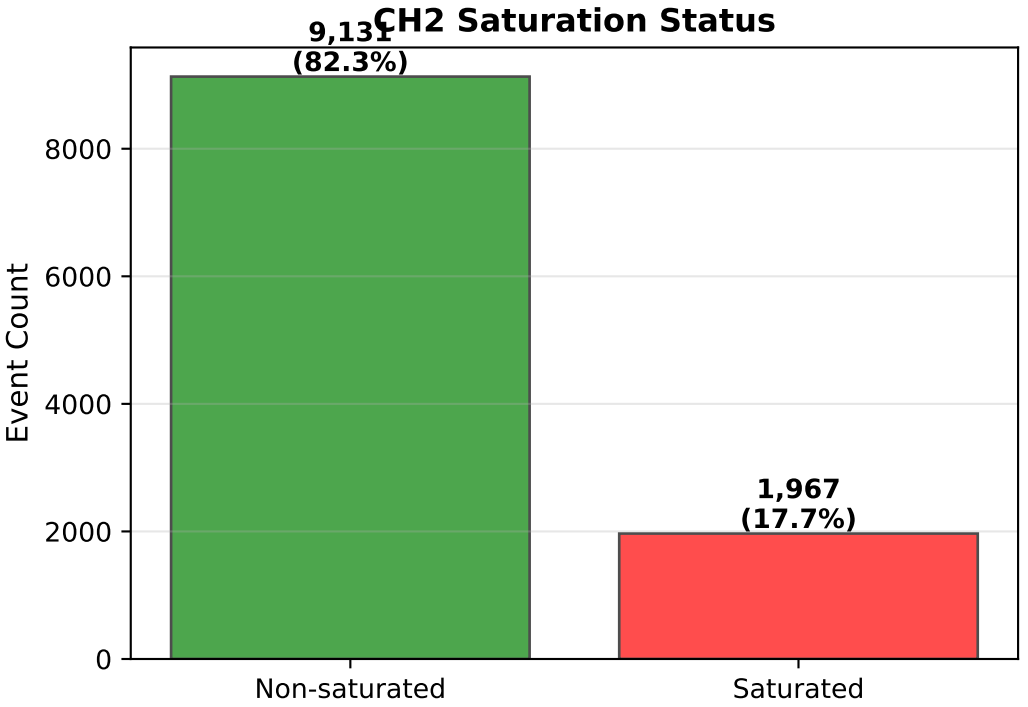
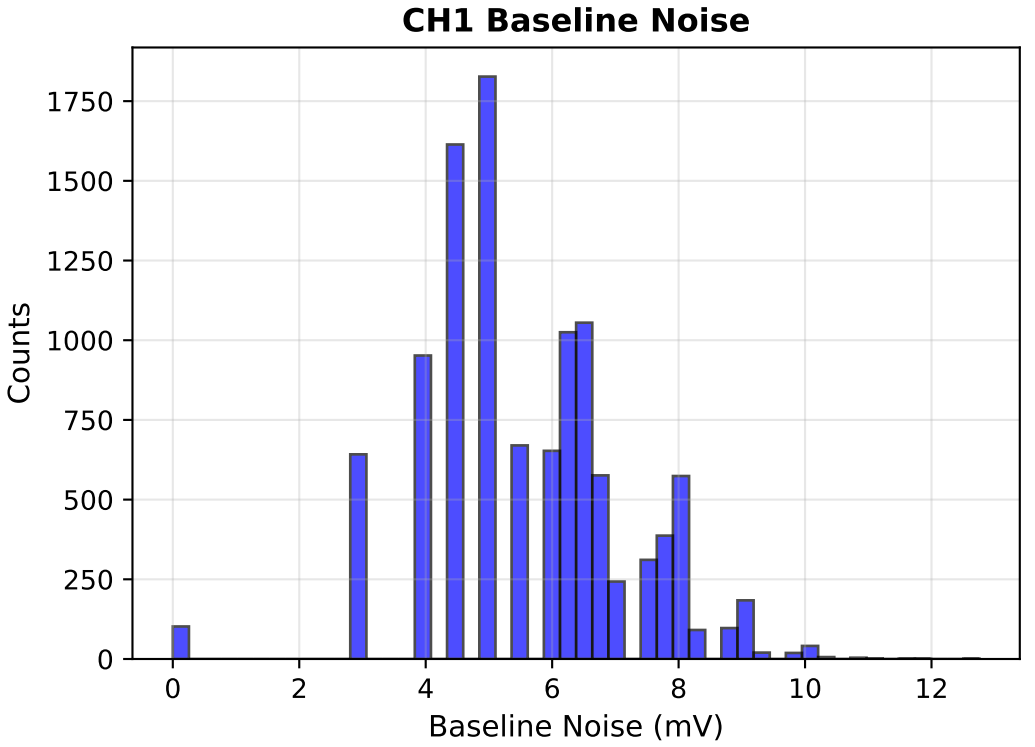
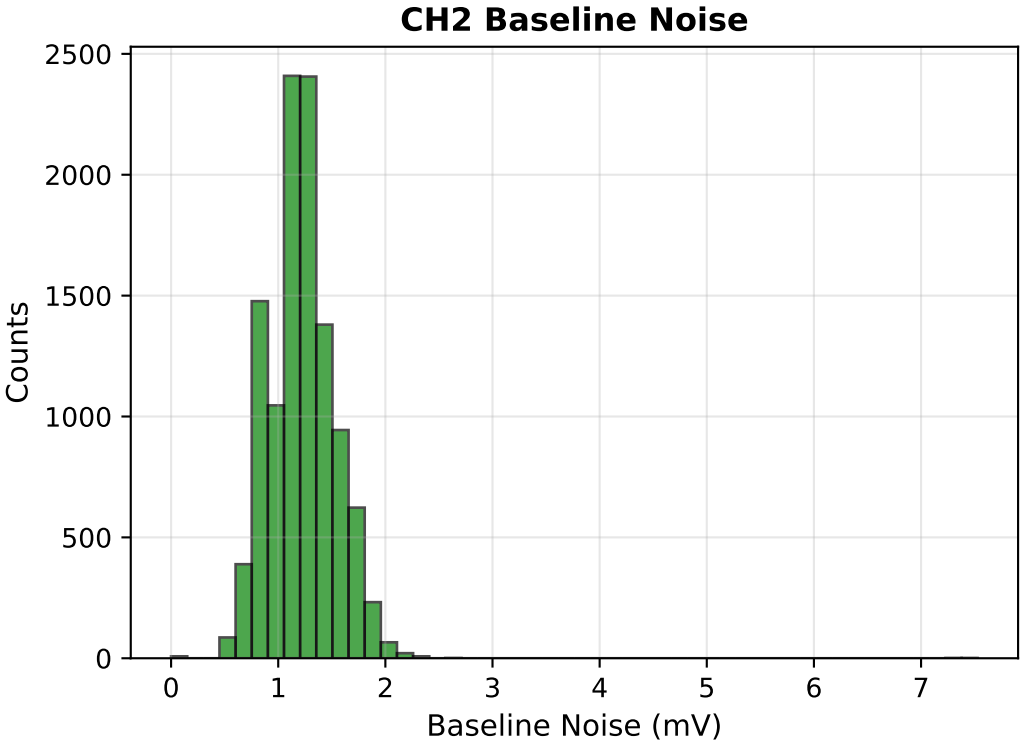
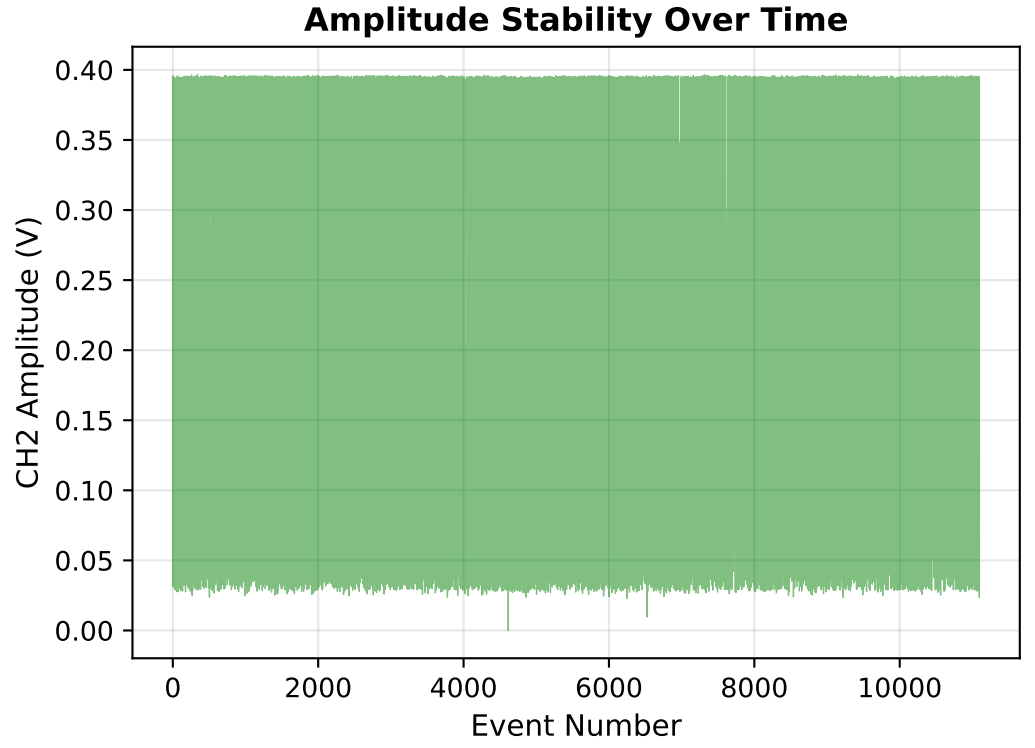








Time-of-Flight Analysis



Event Classification and Quality